

EDUCATION

Texas A&M University, College Station, Texas
Bachelor of Science in Interdisciplinary Engineering
Focus: Biomedical Engineering and Computer Science

Expected Graduation: May 2028
GPA: 3.71

- Engineering Honors (Top 10% of Engineering cohort)
- Timothy J. Holloman Scholarship in Engineering, Texas A&M University
- J. Malon Southerland Aggie Leader Scholarship, Texas A&M University

EXPERIENCE

Tabor Research Group

May 2025 – Present

Undergraduate Researcher

- Developed a Python and RDKit-based computational screening pipeline to filter over **1,000,000,000+ molecules** with **SMARTS-based structural screening**, reducing manual effort by **over 90%** and collecting valid structural candidates for downstream analysis
- **Automated** conformer generation, geometry optimization, and energy analysis for **100,000+ photoacid candidates** across **6 molecular forms** using RDKit, CREST, AIMNet2, Gaussian 16, SLURM, and HPC **supercomputers** including Texas A&M's **Grace** and **VISION (NVIDIA)**
- Analyzed **5+ candidate-selection metrics**, including relative energies, heavy atom counts, closed-shift geometries, and pKa predictions using **pre-trained machine learning models** to prioritize candidates for downstream TD-DFT validation
- Delivered **5+** slide-based research presentations to a lab group of **15+** graduate students, postdocs, and Professor Tabor, effectively communicating screening results and candidate selection

Society of Asian Scientists and Engineers (SASE)

August 2024 – Present

Science Director

- Elected Science Director for TAMU's SASE chapter, sponsored by companies such as Chevron, Boeing, Capital One, and Lockheed Martin, to oversee the planning of **5+ science-related events** per semester for **200+ members**
- Managed an allocated budget and directed a **14-member** committee to organize engaging events including research workshops, outreach events, and academic fair representation to promote undergraduate involvement in research

PROJECTS

Personal Portfolio

May 2026 – July 2026

- Developed a personal portfolio website using Next.js, TypeScript, Tailwind CSS, Framer Motion, and Spline to showcase projects, research experience, and about myself both interactively and dynamically
- Built reusable components and mobile-friendly layouts across **4+ pages**, improving visual consistency

Strive (ACC)

September 2024 – May 2025

Project Member

- Collaborated with a 17-member team to develop Strive, a mobile app for class scheduling built with JavaScript, React Native, and Firebase
- Contributed to frontend development, supported UI design through Figma, and provided usability feedback to enhance app functionality and user experience

TECHNICAL SKILLS

Languages: Python, C++, HTML, JavaScript, CSS, SMILES/SMARTS

Frameworks & Libraries: React, Next.js, React Native, Tailwind CSS, RDKit, pandas, NumPy, SciPy, matplotlib, PyTorch

Software: Gaussian 16, CREST, ChemDraw, SOLIDWORKS, AutoCAD 2D/3D, Onshape, Figma, Excel

Developer Tools: Git/GitHub, VS Code, PyCharm, Sanity CMS, SLURM, High-Performance Computing

Relevant Coursework: Data Structures & Algorithms, Biomedical Computing & Design, Program Design & Concepts, Discrete Structures for Computing, Biostatistics & Data Visualization, Organic Chemistry I & II, Physiology for Bioengineers